

TRAFFICVISION AI

Application Workflows

Core end-to-end process flows across the platform

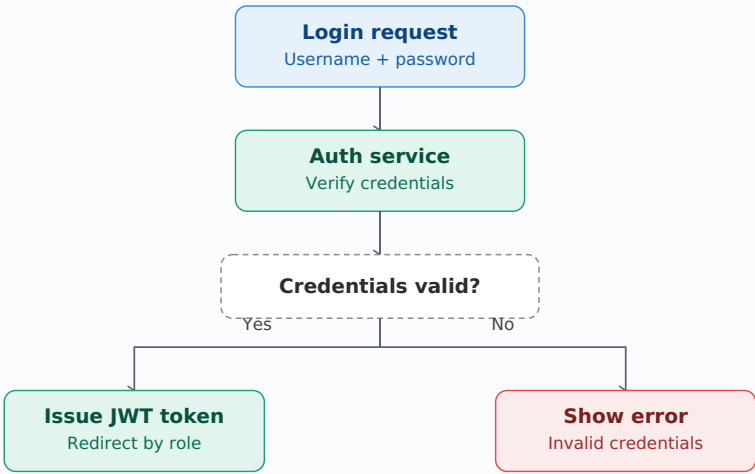


Login & Authentication · Live Monitoring & Alerts · AI Congestion Prediction · Route Recommendation

Milestone 1 — Design & Planning

Login & Authentication

Every session starts here. A user submits credentials, the auth service verifies them against the database, and the flow branches on whether the credentials are valid — this is the entry point that every role-based screen (Admin, Operator, Analyst, Public) depends on.



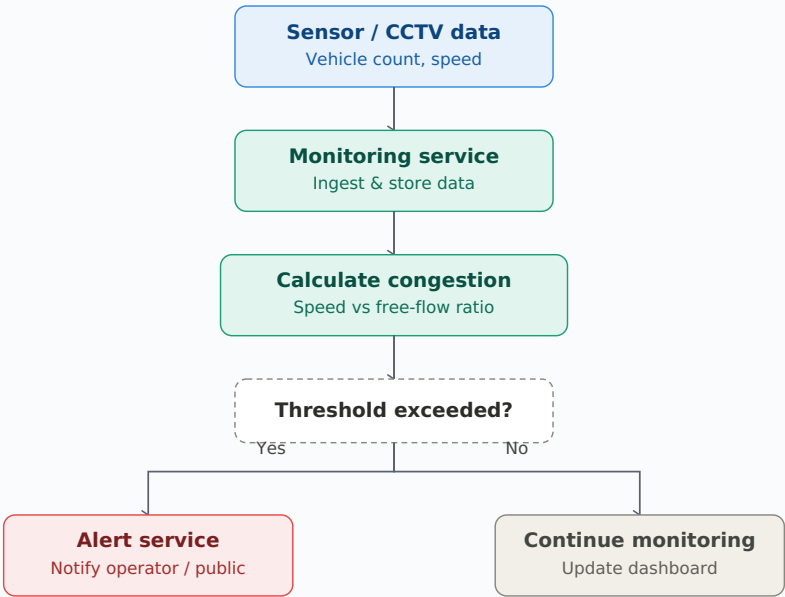
Step-by-step

- **Login request** — user submits username/email + password from the login page.
- **Auth service** — backend looks up the user and verifies the password hash (bcrypt).
- **Decision: Credentials valid?** — the branch point that determines the next step.
- **If yes** — a JWT access token is issued, and the user is redirected to the dashboard matching their role.
- **If no** — an error is returned and shown on the login screen; no token is issued.

Primary service	Auth Service (FastAPI route: /auth/login)
Data touched	USERS table (PostgreSQL)
Security note	Role is never chosen by the user at login — it's read from the stored user record, not the request.

Live Traffic Monitoring & Alerts

This is the reactive, real-time loop that runs continuously once the platform is live. Sensor and CCTV data streams in, congestion is calculated on the fly, and if it crosses a threshold, the alert service fires immediately — otherwise the dashboard simply keeps updating.



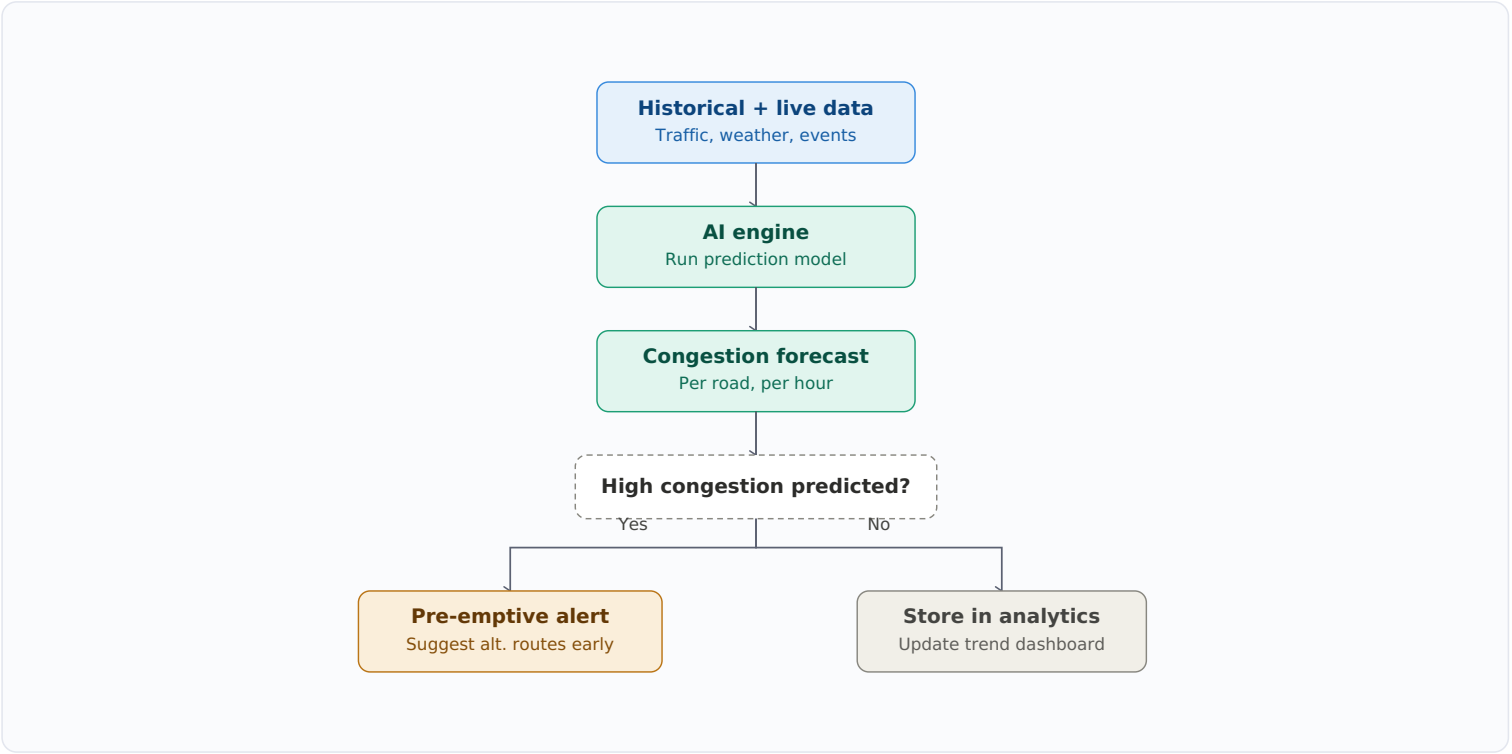
Step-by-step

- **Sensor / CCTV data** — vehicle count and average speed arrive from road sensors or camera feeds.
- **Monitoring service** — ingests the incoming reading and stores it against the relevant road.
- **Calculate congestion** — congestion level is derived from current speed vs. free-flow speed for that road.
- **Decision: Threshold exceeded?** — compares the calculated congestion against a configured alert threshold.
- **If yes** — the alert service creates a notification and pushes it to operators and, where relevant, the public view.
- **If no** — monitoring simply continues and the live dashboard is updated with the latest reading.

Primary services	Traffic Monitoring Service, Alert Service
Data touched	TRAFFIC_DATA, ALERTS tables
Runs	Continuously, triggered by each incoming sensor reading

AI Congestion Prediction

Where Workflow 2 reacts to congestion that has already happened, this workflow looks ahead. Historical and live data feed a trained ML model, which forecasts congestion per road and hour — letting the platform warn people before a jam actually forms.



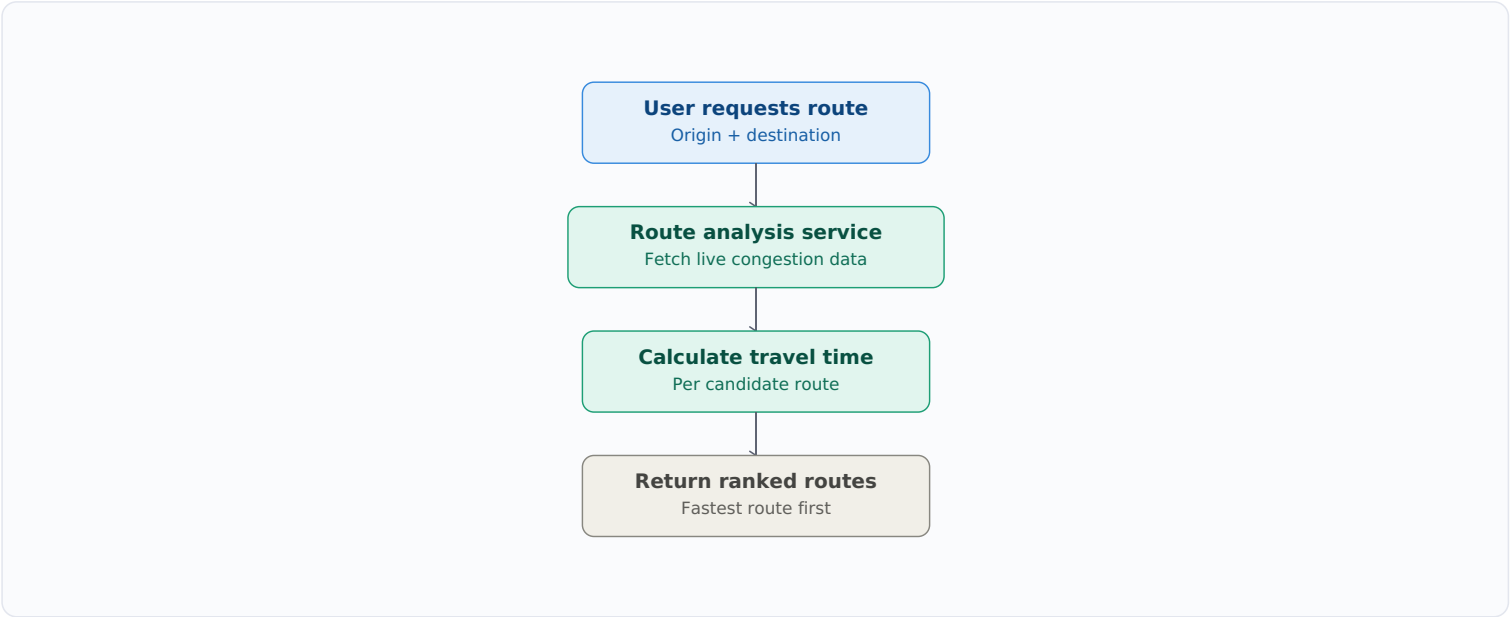
Step-by-step

- **Historical + live data** — past traffic patterns combined with current conditions (weather, events, time of day).
- **AI engine** — runs the trained prediction model (e.g. Random Forest / time-series) on the combined dataset.
- **Congestion forecast** — produces a predicted congestion level for each road, for upcoming time windows.
- **Decision: High congestion predicted?** — checks whether the forecast crosses a "likely to jam" threshold.
- **If yes** — a pre-emptive alert goes out early, along with suggested alternate routes, ahead of the actual congestion.
- **If no** — the forecast is simply stored and surfaces later in the analytics/trend dashboard.

Primary service	AI Engine Service (Scikit-learn / TensorFlow model)
Data touched	PREDICTIONS table, TRAFFIC_DATA (read)
Key difference from Workflow 2	Predictive (before congestion forms) vs. reactive (after it's detected)

Route Recommendation

This is the commuter-facing, on-demand flow — triggered whenever a user asks for directions. Unlike the previous two workflows, there's no branching decision here: every request is simply processed and ranked routes are returned.



Step-by-step

- **User requests route** — user provides an origin and a destination.
- **Route analysis service** — fetches candidate routes between the two points and pulls live congestion data for each.
- **Calculate travel time** — estimates travel time for every candidate route using current conditions.
- **Return ranked routes** — sends the routes back to the user, fastest option listed first.

Primary service	Route Analysis Service
Data touched	ROADS, ROUTES, ROUTE_ROADS, TRAFFIC_DATA (read)
External dependency	OpenStreetMap / Google Maps API for base route candidates